

Lec 11-2, Nov 19, 2015.

- Numerical integration (8.7)

- Over view
- Trapezoidal rule
- Simpson's rule
- Error bounds

- Improper integrals (8.8)

- type I : unbounded intervals
- type II : discontinuous integrands (including unbounded ones)
- Convergence tests

• Numerical Integration

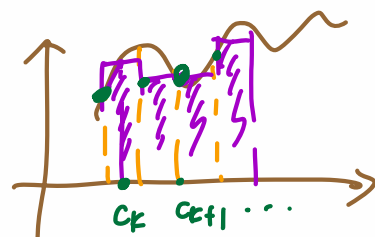
- ▶ Although the Fundamental Theorem of Calculus gives us the ability to compute $\int_a^b f(x) dx$ by finding $F(x)$, it is not always possible to find a closed form of $F(x)$.
- ▶ In particular, some elementary functions, such as

$$f(x) := \frac{\sin x}{x} dx \text{ and } f(x) := e^{x^2} dx = \underline{F(x)} + C$$

do not have an elementary antiderivative.

$x^r, \log, \exp, \sin, \arctan, \dots$

- In practice, approximations are often used.
- For integration approximations, the following approach could be taken:
 - Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$, where $\Delta x_k = \frac{b-a}{n}$, $\forall k$.
 - Then $\int_a^b f(x) dx = \sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx$.
 - Approximate each $\int_{x_{k-1}}^{x_k} f(x) dx$.
 - In a Riemann sum approximation,



$$\int_{x_{k-1}}^{x_k} f(x) dx \approx f(c_k) \cdot \Delta x_k, \text{ (rectangle)}$$

for some $c_k \in \underline{x_{k-1}}, x_k$.

- Directly from Riemann sum approximations, we have
 - left-hand rule ; $\forall k, c_k = x_{k-1}$

— right-hand rule ; $[\forall k, c_k = x_k]$

— midpoint rule . $[\forall k, c_k = \frac{x_{k-1} + x_k}{2}]$

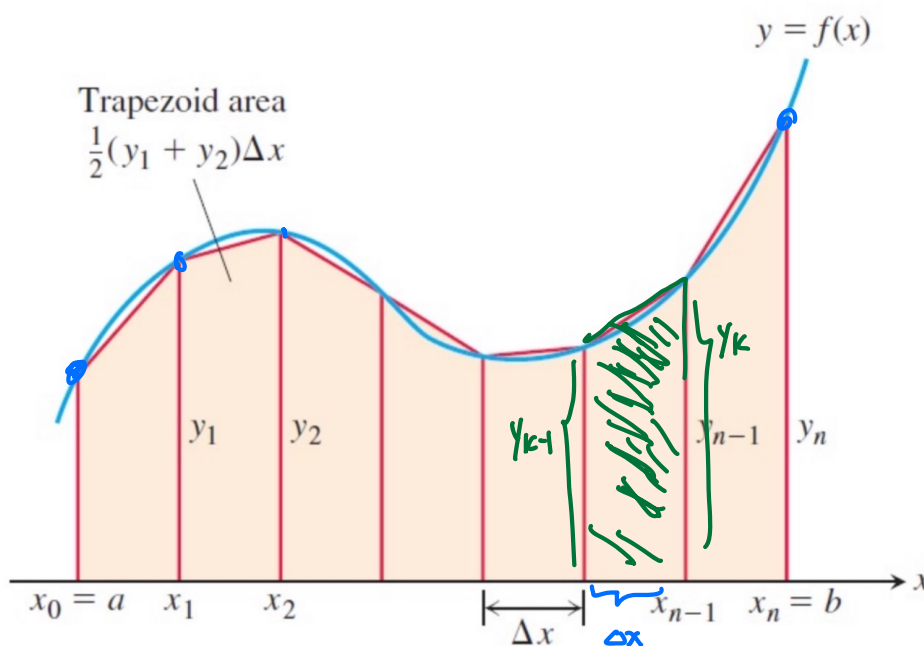


• Today we will introduce two other numerical integration methods.

• Trapezoidal rule

The trapezoidal rule approximates each $\int_{x_{k-1}}^{x_k} f(x) dx$ with the "area" of a trapezoid .

$$\approx \text{Area}(\square) = \frac{1}{2} \Delta x_k (y_{k-1} + y_k)$$



More specifically, instead of $f(c_k) \Delta x_k$ in the Midpoint Rule, we take

$$T_k := \frac{(y_{k-1} + y_k) \Delta x}{2},$$

where $y_j := f(x_j)$. (Note that T_k is the area of a trapezoid if f is nonnegative.)

$$\int_a^b f(x) dx \approx \sum_{k=1}^n T_k = \frac{\Delta x}{2} (y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n),$$

where $\Delta x := (b - a)/n$.

$$\int_a^b f(x) dx = \sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx$$

$$\approx \sum_{k=1}^n T_k$$

$$= \sum_{k=1}^n \frac{y_{k-1} + y_k}{2} \Delta x_k$$

$$= \sum_{k=1}^n \frac{f(x_{k-1}) + f(x_k)}{2} \cdot \Delta x_k = \frac{\Delta x}{2} \left[(f(x_0) + f(x_1)) + (f(x_1) + f(x_2)) + \dots + (f(x_{n-1}) + f(x_n)) \right]$$

$$= \frac{\Delta x}{2} [1 \cdot y_0 + 2 \cdot y_1 + 2 \cdot y_2 + \dots + 2 \cdot y_{n-1} + 1 \cdot y_n]$$

How good are these approximation methods? Let us define

$$\text{error} \triangleq E \triangleq \text{approximated value} - \int_a^b f(x) dx$$

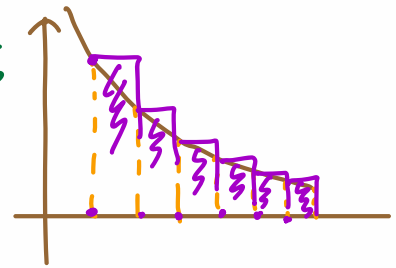
We would like to look at $|E|$.

e.g. Consider $I \triangleq \int_1^{1.6} \frac{1}{x} dx = \ln x \Big|_1^{1.6} = \ln 1.6$

With $\Delta x = 0.1$, use different rules to approximate I .

$$x = 1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6$$

$$f(x) = \frac{1}{1}, \frac{1}{1.1}, \dots, \frac{1}{1.5}, \frac{1}{1.6}$$



Sol¹:
 • Let $f(x) = \frac{1}{x}$.
 • Left-hand rule:

$$\begin{aligned} \sum_{k=1}^n f(x_{k-1}) \cdot \Delta x_k &= [f(1.0) + f(1.1) + \dots + f(1.4) + f(1.5)] \cdot \Delta x_k^{(0.1)} \\ &= \left[\frac{1}{1.0} + \frac{1}{1.1} + \dots + \frac{1}{1.4} + \frac{1}{1.5} \right] \cdot 0.1 \\ &= 0.489260 \dots \end{aligned}$$

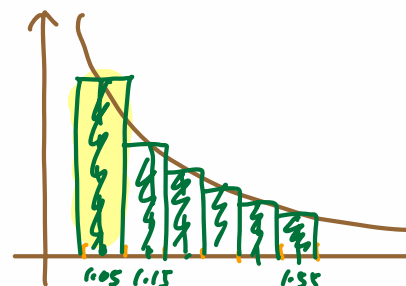
• Right-hand rule:

$$\begin{aligned} \sum_{k=1}^n f(x_k) \Delta x_k &= [f(1.1) + f(1.2) + \dots + f(1.6)] \cdot 0.1 \\ &= \left[\frac{1}{1.1} + \frac{1}{1.2} + \dots + \frac{1}{1.6} \right] \cdot 0.1 \\ &= 0.451760 \dots \end{aligned}$$



• midpoint rule:

$$\begin{aligned} \sum_{k=1}^n f\left(\frac{x_k + x_{k-1}}{2}\right) \Delta x_k &= [f(1.05) + f(1.15) + f(1.25) + f(1.35) + f(1.45) + f(1.55)] \cdot 0.1 \\ &= 0.469750 \dots \end{aligned}$$



• trapezoid rule:

$$\begin{aligned} \sum_{k=1}^n \frac{f(x_{k-1}) + f(x_k)}{2} \cdot \Delta x_k &= \frac{0.1}{2} [1f(1) + 2f(1.1) + 2f(1.2) + 2f(1.3) + 2f(1.4) + 2f(1.5) + 1f(1.6)] \\ &= \frac{0.1}{2} \left[\frac{1}{1} + 2 \cdot \frac{1}{1.1} + 2 \cdot \frac{1}{1.2} + 2 \cdot \frac{1}{1.3} + \dots + \frac{2}{1.5} + \frac{1}{1.6} \right] \\ &= 0.470510 \dots \end{aligned}$$

The true value is

$$I = \ln 1.6 = 0.470003 \dots$$

In the example above, if we consider the errors E_L , E_R , E_M and E_T , then

$$|E_L| = 0.019203,$$

$$|E_R| = 0.018243,$$

$$|E_M| = 0.000253,$$

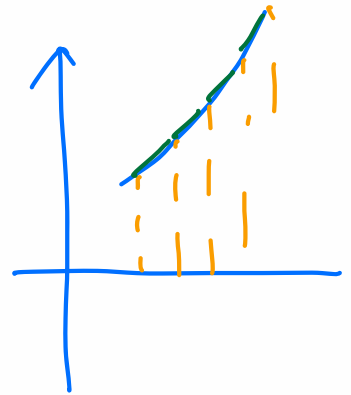
$$|E_T| = 0.000507.$$

$$= 2.5 \times 10^{-4}$$

$$= 5 \times 10^{-4}$$

$\Rightarrow$

$$|E_M| \approx |E_T| < |E_R| \approx |E_L|.$$

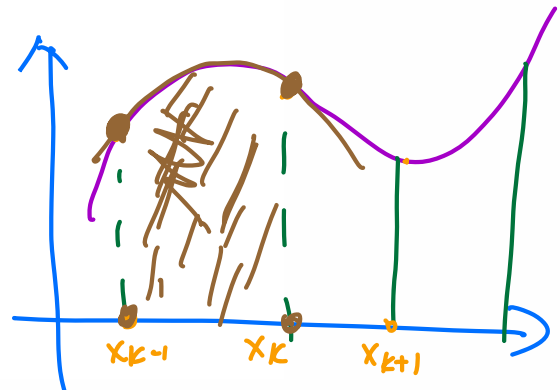
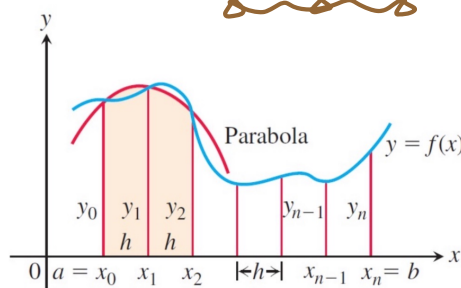
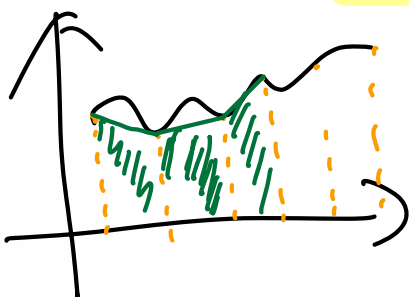


We will see that these are consistent with the error bounds later.

Simpson's Rule

any Riemann sum approximation

- ▶ In the ~~Midpoint Rule~~, f is approximated by a constant $f(c_k)$ over the interval $[x_{k-1}, x_k]$.
- ▶ In the Trapezoidal Rule, over $[x_{k-1}, x_k]$, the function f is approximated by a linear function $y = Ax + B$ whose graph passes through the points (x_{k-1}, y_{k-1}) and (x_k, y_k) .
- ▶ We can take this approach a step further by approximating f with quadratic polynomials $Ax^2 + Bx + C$.



Idea:

- ▶ Consider an evenly spaced partition $P := \{x_0, x_1, \dots, x_n\}$ of $[a, b]$, where n is even.
- ▶ For $k \in \{1, 3, 5, \dots, n-1\}$, over the interval $[x_{k-1}, x_{k+1}]$, consider approximating f by the quadratic function $p_k(x) := A_k x^2 + B_k x + C_k$ whose graph passes through the three points (x_{k-1}, y_{k-1}) , (x_k, y_k) and (x_{k+1}, y_{k+1}) .
- ▶ This approximation is called **Simpson's Rule**, which can be stated as

$$= \int_{x_0}^{x_n} f(x) dx = \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \dots$$

$$\int_a^b f(x) dx \approx \sum_{k \in \{1, 3, \dots, n-1\}} \int_{x_{k-1}}^{x_{k+1}} p_k(x) dx.$$

Aim:

- ▶ We will derive an explicit formula for this approximation.

So!: Step 1: $\int_{x_0}^{x_2} p_1(x) dx$!

We first look at $\int_{x_0}^{x_2} p_1(x) dx$.

Step 2: Shift!

To compute it, we may shift the graph

so that $x_0 = -\Delta x$, $x_1 = 0$ and $x_2 = \Delta x$, then compute the integral.

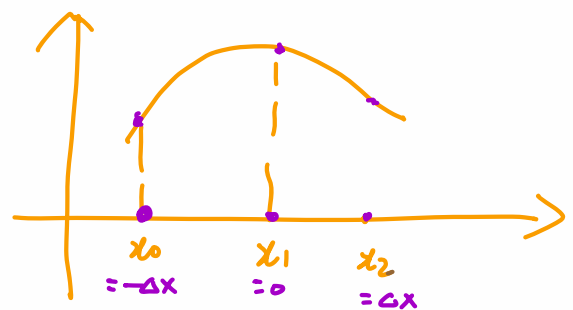
[Because horizontal shifts do not change the value of an integral.]

$$\int_{x_0}^{x_2} A x^2 + B x + C dx = ?? \text{ s.t. } \begin{cases} A x_0^2 + B x_0 + C = y_0 \\ A x_1^2 + \dots = y_1 \\ A x_2^2 + \dots = y_2 \end{cases}$$

Step 3: Integrate an easier shifted function!

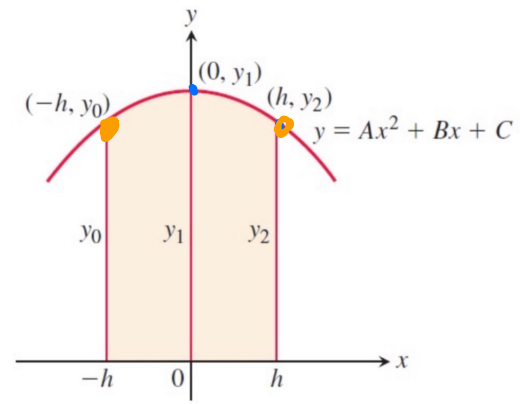
Let $h \triangleq \Delta x$ and write

$$\underline{q_1(x) = A x^2 + B x + C = p_1(x + h + x_0)}$$



Then

$$\begin{aligned}
 \int_{x_0}^{x_2} p_1(u) du &= \int_{x_0-h-x_0}^{x_2-h-x_0} p_1(x+x_0+h) dx \\
 &= \int_{-h}^h p_1(x+x_0+h) dx \\
 &= \int_{-h}^h q_1(x) dx \\
 &= \int_{-h}^h (Ax^2+Bx+C) dx \\
 &= A \frac{1}{3} x^3 \Big|_{-h}^h + C \cdot 2h \\
 &= A \frac{2}{3} h^3 + C \cdot 2h \\
 &= \frac{h}{3} (2Ah^2 + 6C).
 \end{aligned}$$



①

Step 4: Compute coefficients in terms of y -values!

Since $(-h, y_0)$, $(0, y_1)$ and (h, y_2) are all on the graph of

$y = Ax^2 + Bx + C$, we have

$$\begin{cases}
 y_0 = A(-h)^2 + B(-h) + C, \\
 y_1 = A \cdot 0^2 + B \cdot 0 + C \\
 y_2 = A \cdot h^2 + B \cdot h + C.
 \end{cases}$$

$$\Rightarrow \begin{cases} C = y_1 \\ y_0 + y_2 = Ah^2 \cdot 2 + 2C \end{cases}$$

$$\Rightarrow \begin{cases} 6C = 6y_1 \\ 2Ah^2 = y_0 + y_2 - 2y_1. \end{cases}$$

Step 5: Compute the integral in terms of y -values!

By ①, we have

$$\begin{aligned} \int_{x_0}^{x_2} p_1(u) du &= \frac{h}{3} (2Ah^2 + 6C) = \frac{h}{3} (y_0 + y_2 - 2y_1 + 6y_1) \\ &= \frac{h}{3} (y_0 + 4y_1 + y_2). \end{aligned}$$

Step 6: $\int_{x_{k-1}}^{x_{k+1}} p_k(x) dx$!

The same argument also leads to

$$\int_{x_{k-1}}^{x_{k+1}} p_k(x) dx = \frac{h}{3} (y_{k-1} + 4y_k + y_{k+1}).$$

Step 7: Combine!

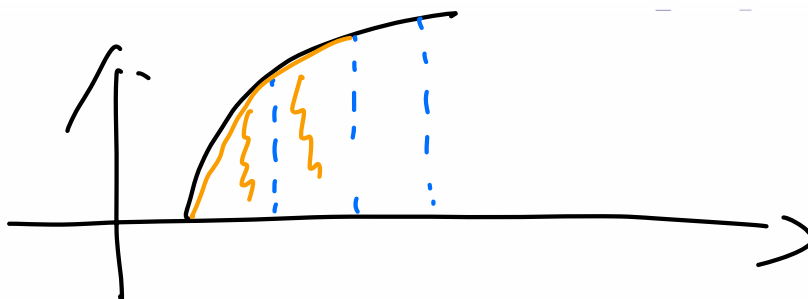
Therefore,

$$\begin{aligned} \int_a^b f(x) dx &\approx \int_{x_0}^{x_2} p_1(x) dx + \int_{x_2}^{x_4} p_3(x) dx + \cdots + \int_{x_{n-2}}^{x_n} p_{n-1}(x) dx \\ &= \int_{x_0}^{x_2} p_1(x) dx + \int_{x_2}^{x_4} p_3(x) dx + \cdots + \int_{x_{n-2}}^{x_n} p_{n-1}(x) dx \\ &= \frac{h}{3} (y_0 + 4y_1 + y_2 + y_2 + 4y_3 + y_4 + y_4 + 4y_5 + y_6 + \cdots + y_{n-2} + 4y_{n-1} + y_n). \end{aligned}$$

Hence, an explicit formula for **Simpson's Rule** is given by

$$\int_a^b f(x) dx \approx \frac{h}{3} (y_0 + 4y_1 + 2y_2 + \cdots + 2y_{n-2} + 4y_{n-1} + y_n),$$

where n is even, and the coefficients are 1, 4, 2, 4, 2, 4, 2, ..., 4, 2, 4, and 1.



e.g. Use Simpson's rule to approximate $I \triangleq \int_1^{1.6} \frac{1}{x} dx$, with $\Delta x = 0.1$, and compare its error with those of other rules. [$f(x) = \frac{1}{x}$].

$$\cdot I \approx \frac{0.1}{3} (1f(1) + 4f(1.1) + 2f(1.2) + 4f(1.3) + 2f(1.4) + 4f(1.5) + 1f(1.6))$$

$$= 0.470006 \dots \checkmark$$

$$\cdot \text{Exact value} = \ln 1.6 = 0.470003 \dots \checkmark$$

$$\cdot |E_S| = \underline{0.000003} \approx 3 \times 10^{-6}$$

$$\cdot |E_S| < |E_M| \left(\approx |E_T| < |E_R| \approx |E_L| \right).$$

Error Bounds

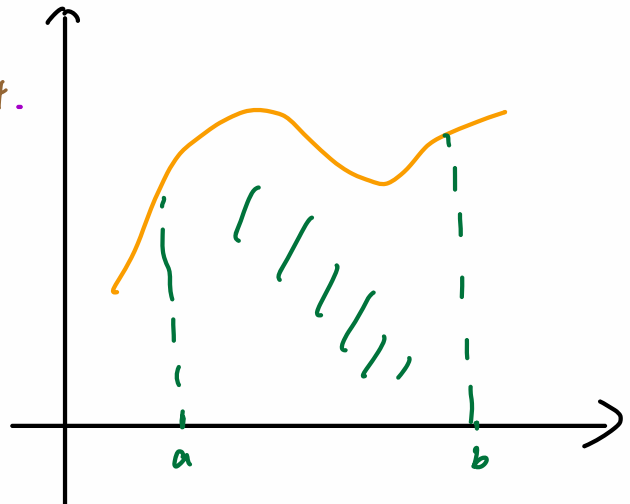
Let f be a function that is integrable on $[a, b]$.

Let E_S, E_T, E_M, E_L and E_R be the error of Simpson's rule, trapezoidal rule, midpoint rule, left-hand rule and right-hand rule for approximating $\int_a^b f(x) dx$, respectively.

$$\text{Let } \max |f^{(i)}| \triangleq \max_{x \in [a, b]} |f^{(i)}(x)|, i=1, 2, 3, 4.$$

$$(\text{e.g. } \max |f^{(2)}| \triangleq \max_{x \in [a, b]} |f''(x)|.)$$

$$E_S \triangleq \text{Simpson's rule value} - \text{true value}$$



We list the error bounds (Optional; will be covered in "Numerical Analysis")

$$\begin{aligned}
 \textcircled{1} \quad |E_L| &\leq \frac{(b-a)^2}{2n} \max |f'|, & |E_R| &\leq \frac{(b-a)^2}{2n} \max |f'|, \\
 |E_T| &\leq \frac{(b-a)^3}{12n^2} \max |f''|, & |E_M| &\leq \frac{(b-a)^3}{24n^2} \max |f''|, \\
 \textcircled{5} \quad |E_S| &\leq \frac{(b-a)^5}{180n^4} \max |f^{(4)}|.
 \end{aligned}$$

no need to memorize

$C \cdot \frac{1}{n}$
 $C' \cdot \frac{1}{n^2}$
 $C_2 \cdot \frac{1}{n^2}$
 $C_3 \cdot \frac{1}{n^4}$

Generally speaking, from ^{the} most accurate to the least (important!)

Simpson > midpoint $\approx$ trapezoid > left $\approx$ right.

$$O\left(\frac{1}{n^4}\right)$$

$$O\left(\frac{1}{n^2}\right)$$

$$O\left(\frac{1}{n}\right)$$

e.g. If we want to approximate $\int_0^1 e^{x^2} dx$ with error

having absolute value less than 10^{-5} , how many sub-intervals do we need at least? Give the answer for Simpson's rule and the trapezoid rule.

Sol¹: Let $f(x) = e^{x^2}$.

$$\therefore f'(x) = e^{x^2} \cdot 2x$$

$$f''(x) = e^{x^2} (4x^2 + 2).$$

$$f'''(x) = e^{x^2} (8x^3 + 12x).$$

$$f^{(4)}(x) = e^{x^2} (16x^4 + 48x^2 + 12).$$

$$\therefore \max |f''| = \max_{x \in [0,1]} f''(x) = f''(1) = 6e,$$

$$\max |f^{(4)}| = f^{(4)}(1) = 76e.$$

∴ For Simpson's rule, we want

$$\frac{(b-a)^5}{180n^4} \max |f^{(4)}| = \frac{(1-0)^5}{180n^4} 76e < 10^{-5}$$

$$\Rightarrow 180n^4 > 76 \cdot 10^5 e$$

$$\Rightarrow n \geq 19$$

∴ For the trapezoidal rule, we want

$$\frac{(b-a)^3}{12n^2} \max |f''| = \frac{1^3}{12n^2} 6e < 10^{-5}$$

$$\Rightarrow 12n^2 > 10^5 \cdot 6e$$

$$\Rightarrow n \geq 369$$

□

Improper integrals

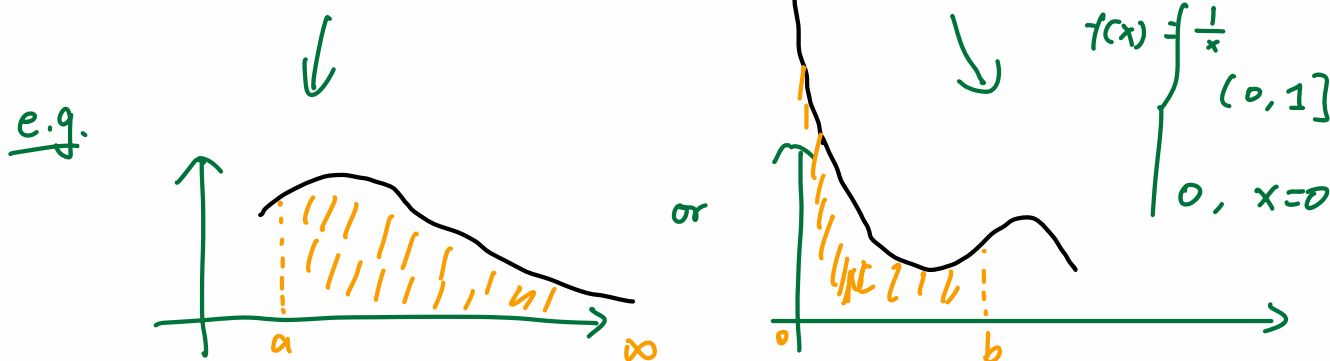
Definite integrals $\int_a^b f(x) dx$ assume two conditions:

- Interval $[a, b]$ is bounded, and; $[1, 3], [-\pi, \frac{3\pi}{2}]$
- f is bounded on $[a, b]$ (otherwise not Riemann integrable).

Hence, Riemann integrals cannot handle functions over unbounded intervals or those that have unbounded ranges.

We introduce improper integrals which are limits of Riemann integrals.

These will cover unbounded domains or unbounded integrands.



In probability (statistics, financial mathematics), these integrals are fundamental. (§8.9; STA2001;)

(pdf)

— A probability density function is a certain nonnegative function f that satisfies

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

— For discrete case, $f(x_1) + f(x_2) + \dots + f(x_n) = 1$, where $f(x_i)$ are probabilities.

Definition

- ▶ Let $a \in \mathbb{R}$. If f is integrable on $[a, b]$ for every $b \in [a, \infty)$, then

$$\int_a^{\infty} f(x) dx := \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

- ▶ Let $b \in \mathbb{R}$. If f is integrable on $[a, b]$ for every $a \in (-\infty, b]$, then

$$\int_{-\infty}^b f(x) dx := \lim_{a \rightarrow -\infty} \int_a^b f(x) dx.$$

Definition

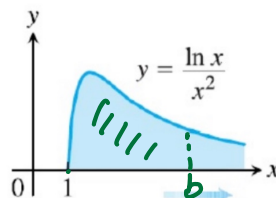
- ▶ An improper integral is said to be **convergent** if the corresponding limit exists, and is said to be **divergent** if the limit does not exist (as a real number). The limit $\pm\infty$ also means divergent.
- ▶ We define

$$\int_{-\infty}^{\infty} f(x) dx := \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx$$

whenever both improper integrals on the right converge. We may use any real number c in this definition.

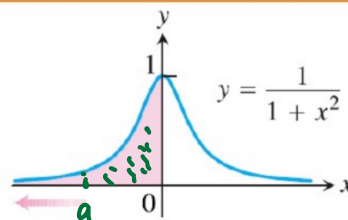
1. Upper limit

$$\int_1^{\infty} \frac{\ln x}{x^2} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{\ln x}{x^2} dx$$



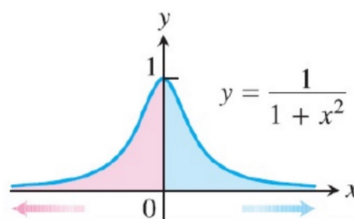
2. Lower limit

$$\int_{-\infty}^0 \frac{dx}{1+x^2} = \lim_{a \rightarrow -\infty} \int_a^0 \frac{dx}{1+x^2}$$



3. Both limits

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \lim_{b \rightarrow -\infty} \int_b^0 \frac{dx}{1+x^2} + \lim_{c \rightarrow \infty} \int_0^c \frac{dx}{1+x^2}$$



Rmk. Our definition of $\int_{-\infty}^{\infty}$ is NOT the same as $\lim_{a \rightarrow \infty} \int_{-a}^a$.

(Why?)

e.g. $f(x) = \begin{cases} 1, & x \geq 1 \\ x, & x \in (-1, 1) \\ -1, & x \leq -1. \end{cases}$

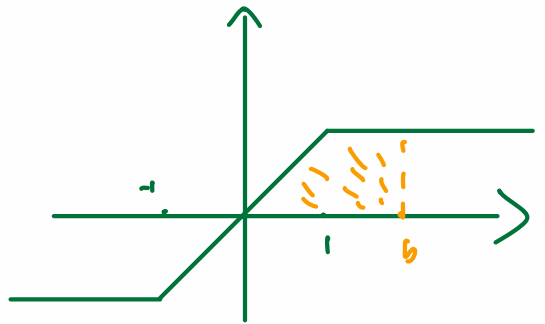
$$\int_{-\infty}^{\infty} f(x) dx$$

$$\lim_{a \rightarrow \infty} \int_{-a}^a f(x) dx$$

$$\bullet \int_0^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_0^b f(x) dx = \lim_{b \rightarrow \infty} \left[\int_0^1 + \int_1^b \right] = \lim_{b \rightarrow \infty} \left[\frac{1}{2} + (b-1) \right] = \infty.$$

$$\bullet \int_{-\infty}^0 f(x) dx = \lim_{a \rightarrow -\infty} \int_a^0 f(x) dx = -\infty$$

$\Rightarrow \int_{-\infty}^{\infty} f(x) dx$ does not exist.



$$\begin{aligned} \bullet \lim_{a \rightarrow \infty} \int_{-a}^a f(x) dx &= \lim_{a \rightarrow \infty} \left[\int_0^a f dx + \int_{-a}^0 f dx \right] \\ &= \lim_{a \rightarrow \infty} 0 \\ &= 0. \end{aligned}$$

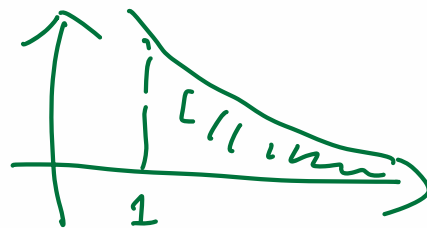
$$\therefore \int_{-\infty}^{\infty} f(x) dx \neq \lim_{a \rightarrow \infty} \int_{-a}^a f(x) dx$$

□

e.g. (a) $\int_1^{\infty} \frac{1}{x} dx = ?$

Sol': LHS = $\lim_{b \rightarrow \infty} \int_1^b \frac{1}{x} dx = \lim_{b \rightarrow \infty} \ln|x| \Big|_1^b = \lim_{b \rightarrow \infty} (\ln b) = \infty$.

$\therefore$ LHS is divergent.



(b) $\int_1^{\infty} x^{\alpha} dx, \alpha \neq -1$.

Sol': LHS = $\lim_{b \rightarrow \infty} \int_1^b x^{\alpha} dx = \lim_{b \rightarrow \infty} \left(\frac{1}{\alpha+1} x^{\alpha+1} \Big|_1^b \right)$

$= \lim_{b \rightarrow \infty} \left(\frac{1}{\alpha+1} (b^{\alpha+1} - 1) \right)$

$\alpha = 2, 1, 0$

$\frac{1}{-10+1} b^{-10+1} \rightarrow 0$

$\alpha = -10, -2 \quad (b \rightarrow \infty)$

• if $\alpha+1 < 0$, we have $\lim_{b \rightarrow \infty} b^{\alpha+1} = 0$

$\Rightarrow$ LHS = $-\frac{1}{\alpha+1}$ (convergent).

• if $\alpha+1 > 0$, we have $\lim_{b \rightarrow \infty} b^{\alpha+1} = \infty$

$\Rightarrow$ LHS = ∞ (divergent) !
(2)

Considering (a), we have

$$\int_1^{\infty} x^{\alpha} dx = \begin{cases} -\frac{1}{\alpha+1}, & \text{if } \alpha < -1 ; \\ \text{divergent} & \text{if } \alpha \geq -1 . \\ \quad (c = \infty) \end{cases}$$

An equivalent form is given by

$$\int_1^{\infty} \frac{1}{x^p} dx = \begin{cases} \frac{1}{p-1}, & \text{if } p > 1; \text{ (convergent)} \\ \frac{1}{1-p} x^{1-p} \Big|_1^{\infty} = \begin{cases} \infty, & \text{if } p \leq 1. \text{ (divergent)} \\ 0 - \frac{1}{1-p}, & p > 1 \end{cases} \end{cases}$$

$$(c) \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$$

$$\begin{aligned} \text{Sol}^1: \int_0^{\infty} \frac{1}{1+x^2} dx &= \lim_{b \rightarrow \infty} \int_0^b \frac{1}{1+x^2} dx = \lim_{b \rightarrow \infty} (\arctan x \Big|_0^b) \\ &= \lim_{b \rightarrow \infty} (\arctan b - 0) = \frac{\pi}{2} \end{aligned}$$

$$\therefore \int_0^{\infty} \frac{1}{1+x^2} dx \text{ converges (to } \frac{\pi}{2} \text{)}$$

$$\begin{aligned} \int_{-\infty}^0 \frac{1}{1+x^2} dx &= \lim_{a \rightarrow -\infty} \int_a^0 \frac{1}{1+x^2} dx = \lim_{a \rightarrow -\infty} (\arctan x \Big|_a^0) = \lim_{a \rightarrow -\infty} (\arctan 0 - \arctan a) \\ &= -(-\frac{\pi}{2}) = \frac{\pi}{2}. \end{aligned}$$

$$\therefore \int_{-\infty}^0 \frac{1}{1+x^2} dx \text{ also converges (to } \frac{\pi}{2} \text{)}.$$

$$\therefore \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx$$

$$\begin{aligned} &= \frac{\pi}{2} + \frac{\pi}{2} \\ &= \pi. \end{aligned}$$

Type II

The second type of improper integrals involves discontinuous integrands.

In particular, it covers cases like $f(x) = \frac{1}{x}$ which has an infinite discontinuity at $x = 0$.

Improper Integrals of Type II

The second type of improper integrals involves discontinuous integrands.

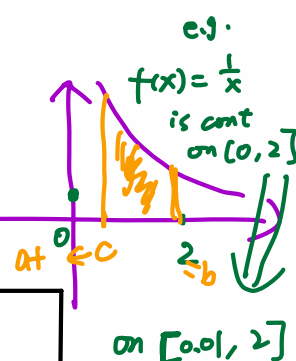
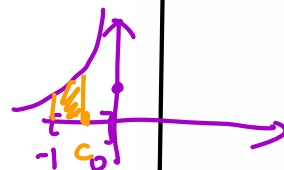
Definition

- If f is continuous on $[a, b)$ and is discontinuous at b , then

$$\int_a^b f(x) dx := \lim_{c \rightarrow b^-} \int_a^c f(x) dx.$$

- If f is continuous on $(a, b]$ and is discontinuous at a , then

$$\int_a^b f(x) dx := \lim_{c \rightarrow a^+} \int_c^b f(x) dx.$$



Definition

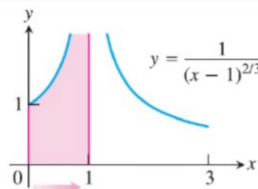
- If f is discontinuous at c , where $a < c < b$, and is continuous on $[a, b] \setminus \{c\}$, then

$$\int_a^b f(x) dx := \int_a^c f(x) dx + \int_c^b f(x) dx,$$

provided that both improper integrals on the right converge.

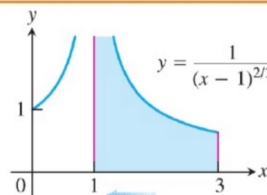
4. Upper endpoint

$$\int_0^1 \frac{dx}{(x-1)^{2/3}} = \lim_{b \rightarrow 1^-} \int_0^b \frac{dx}{(x-1)^{2/3}}$$



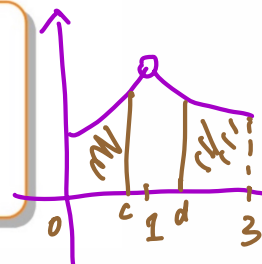
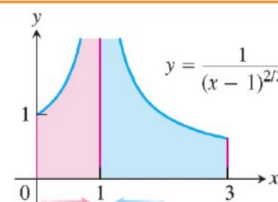
5. Lower endpoint

$$\int_1^3 \frac{dx}{(x-1)^{2/3}} = \lim_{d \rightarrow 1^+} \int_d^3 \frac{dx}{(x-1)^{2/3}}$$



6. Interior point

$$\int_0^3 \frac{dx}{(x-1)^{2/3}} = \int_0^1 \frac{dx}{(x-1)^{2/3}} + \int_1^3 \frac{dx}{(x-1)^{2/3}}$$



Rmks

- If f is continuous on $[a, b)$ but $\lim_{x \rightarrow b^-} f(x) \in \{-\infty, \infty\}$, then

$\int_a^b f(x) dx$ is NOT a Riemann integral (since the unbounded function is not Riemann integrable), although the notation looks

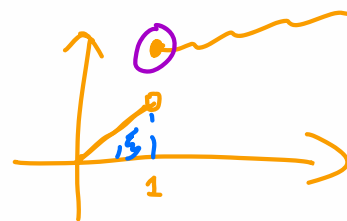
the same . e.g. $\int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}} dx$.

- If f is continuous on $[a, b)$ and f has a jump/removable left-discontinuity at b , then it can be shown that the improper

integral $\int_a^b f(x) dx$ has the same value as the Riemann integral

$\int_a^b f(x) dx$. [Proof omitted].

e.g. $\underline{f(x)} = \begin{cases} x, & \text{if } x \in [0, 1) \\ 2, & \text{if } x = 1. \end{cases}$



Improper integral

$$\int_0^1 f(x) dx$$

$$\stackrel{\text{def}}{=} \lim_{b \rightarrow 1^-} \int_0^b f(x) dx$$

$$= \lim_{b \rightarrow 1^-} \int_0^b x dx$$

$$\stackrel{\text{FTC2}}{=} \lim_{b \rightarrow 1^-} \left(\frac{1}{2} x^2 \right) \Big|_0^b$$

$$= \frac{1}{2} .$$

Riemann integral

$$\int_0^1 f(x) dx$$

$$\stackrel{\text{thm}}{=} \int_0^1 g(x) dx$$

$$= \int_0^1 x dx$$

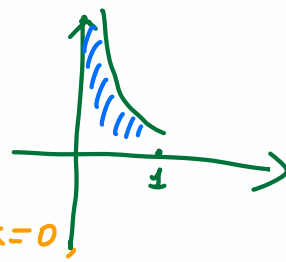
$$\stackrel{\text{FTC}}{=} \left(\frac{1}{2} x^2 \right) \Big|_0^1$$

$$= \frac{1}{2} .$$

$\left[\begin{array}{l} g(x) \triangleq x, x \in [0, 1] \\ g \text{ is the same as } f \text{ except } x=1 \end{array} \right]$

$$\left[G(x) = \frac{1}{2} x^2 \right]$$

e.g. Suppose $p > 0$. $\int_0^1 \frac{1}{x^p} dx = ?$



Sol': $\lim_{x \rightarrow 0^+} \frac{1}{x^p} = \infty$. (f has an essential discontinuity at $x=0$,
and is continuous on $(0, 1]$.)

• If $p=1$,

$$\text{LHS} = \lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{x} dx = \lim_{a \rightarrow 0^+} (\ln x) \Big|_a^1 = \lim_{a \rightarrow 0^+} (0 - \ln a) = \infty.$$

• If $p \neq 1$,

$$\text{LHS} = \lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{x^p} dx = \lim_{a \rightarrow 0^+} \left(\frac{1}{1-p} x^{1-p} \right) \Big|_a^1$$

$$= \lim_{a \rightarrow 0^+} \frac{1}{1-p} (1 - a^{1-p}) = \begin{cases} \frac{1}{1-p}, & \text{if } p \in (0, 1); \\ \infty, & \text{if } p > 1. \end{cases}$$

draft:

$$\left| \begin{array}{l} p=2 \Rightarrow a^{-1} \rightarrow \infty \\ (a \rightarrow 0^+) \end{array} \right.$$

$$\left| \begin{array}{l} p=0.5 \Rightarrow a^{0.5} \rightarrow 0 \\ (a \rightarrow 0^+) \end{array} \right.$$

$\therefore$ Hence,

$$\int_0^1 \frac{1}{x^p} dx \quad \left\{ \begin{array}{l} \text{converges to } \frac{1}{1-p}, \quad p \in (0, 1); \\ \text{diverges (to } \infty), \quad p \geq 1. \end{array} \right.$$

Combined with Type-I improper integrals, this means that:

- If $0 < p < 1$: $\int_0^1 \frac{1}{x^p} dx$ converges, $\int_1^{\infty} \frac{1}{x^p} dx$ diverges.
- If $p > 1$: $\int_0^1 \frac{1}{x^p} dx$ diverges, $\int_1^{\infty} \frac{1}{x^p} dx$ converges.
- If $p = 1$: $\int_0^1 \frac{1}{x^p} dx$ diverges, $\int_1^{\infty} \frac{1}{x^p} dx$ diverges.

Convergence tests.

- consider $\int_0^{\infty} e^{-x^2} dx$. Here $f(x) = e^{-x^2}$ has no elementary antiderivative, so numerical method is required. Before approximating its value, we may want to know whether it converges at all.
- Convergence tests may be used for such purposes.

Theorem (Direct Comparison Test)

Suppose that $a \in \mathbb{R}$, and suppose that f and g are continuous functions on $[a, \infty)$. If there exists $c \in [a, \infty)$ such that $0 \leq f(x) \leq g(x)$ for all $x \in [c, \infty)$, then the following statements hold:

"integrable on $[a, b]$, $\forall b \geq a$ "
is enough.

- (i) If $\int_a^\infty g(x) dx$ converges, then $\int_a^\infty f(x) dx$ converges.
- (ii) If $\int_a^\infty f(x) dx$ diverges, then $\int_a^\infty g(x) dx$ diverges.

Proof idea (not a complete proof).

Step 1. $\int_c^\infty g(x) dx$! (only the tail matters.)

• Note that (i) and (ii) are equivalent. We outline (i).

• Since $\int_a^\infty g(x) dx = \int_a^c g(x) dx + \int_c^\infty g(x) dx$ and

$\int_a^c g(x) dx$ is a definite integral which has finite value,

$\int_a^\infty g(x) dx$ converges if and only if $\int_c^\infty g(x) dx$ converges.

• Similarly, $\int_a^\infty f(x) dx$ converges if and only if $\int_c^\infty f(x) dx$ converges.

• Hence, it suffices to show that if $\int_c^\infty g(x) dx$ converges, then

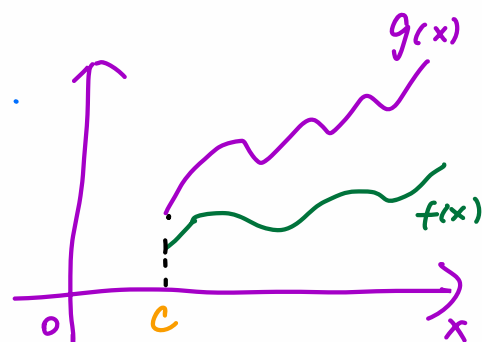
$\int_c^\infty f(x) dx$ converges.

Step 2. Upper bound:

Let $F(t) \triangleq \int_c^t f(x) dx$ and $G(t) \triangleq \int_c^t g(x) dx$, for $t \geq c$.

Since $0 \leq f(x) \leq g(x)$ for all $x \in [c, \infty)$, we have

$0 \leq F(t) \leq G(t)$, $\forall t \geq c$ and F, G are both increasing.

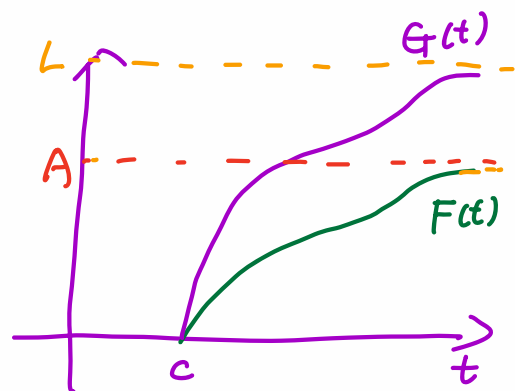


Step 3. (Optional Thm) $F(t)$ increasing and has an upper bound

$\Rightarrow \lim_{t \rightarrow \infty} F(t)$ exists. (*)

Assume that $\int_c^\infty g(x) dx$ converges.

Then $\lim_{t \rightarrow \infty} G(t) = \int_c^\infty g(x) dx = L$ for some $L \in \mathbb{R}$.



$F(t)$ is increasing ($\because F'(t) = f(t) \geq 0$)

$\therefore F(t)$ is bounded above ($F(t) \leq G(t) \leq L$, $\forall t \geq c$) and increasing on (c, ∞)

$\Rightarrow \lim_{t \rightarrow \infty} F(t)$ exists (denoted by A).

$$= \lim_{t \rightarrow \infty} \int_c^t f(x) dx = \int_c^\infty f(x) dx$$

(*) is known as "the monotone bounded convergence theorem",

which is a basic property of real line. This will be covered

in Mathematical Analysis.

Step 4. Conclude!

Then $\lim_{t \rightarrow \infty} F(t) = A$. So $\int_0^{\infty} f(x) dx = A$ (convergent!) $\square$

e.g. Determine whether $\int_0^{\infty} e^{-x^2} dx$ converges or not.

Sol': $\therefore e^{x^2} \geq e^x, \forall x \geq 1.$

$$\therefore 0 \leq e^{-x^2} \leq e^{-x}, \forall x \geq 1.$$

$$\begin{aligned} \therefore \int_0^{\infty} e^{-x} dx &= \left(\lim_{b \rightarrow \infty} \int_0^b e^{-x} dx = -e^{-x} \Big|_0^{\infty} = 0 - (-e^0) \right. \\ &= \lim_{b \rightarrow \infty} (-e^{-x}) \Big|_0^b \\ &= \lim_{b \rightarrow \infty} (-e^{-b} + e^0) \end{aligned} \quad \begin{aligned} &= 1 < \infty \\ &\text{(convergent!)} \end{aligned}$$

$\therefore$ By Direct Comparison test,

we have $\int_0^{\infty} e^{-x^2} dx$ converges.

. 19.

e.g. $\int_2^{\infty} \frac{\sqrt[3]{x^7+2}}{x^3 \ln x} dx$

Sol': $\left[\begin{aligned} \text{Draft: } & \frac{(x^7+2)^{\frac{1}{3}}}{x^3 \ln x} \approx \frac{(x^7)^{\frac{1}{3}}}{x^3 \cdot x^0} = x^{-\frac{2}{3}} \quad (x \rightarrow \infty) \\ & \int_2^{\infty} x^{-\frac{2}{3}} dx = \frac{1}{\frac{1}{3}} x^{\frac{1}{3}} \Big|_2^{\infty} = \infty \\ & \text{Try divergent \& easy-to-integrate functions, e.g. } \frac{1}{x \ln x}. \end{aligned} \right]$

$$\text{integrand} = \frac{(x^2+2)^{\frac{1}{3}}}{x^3 \ln x} = \frac{(x^2+2)^{\frac{1}{3}}}{x^2} \cdot \frac{1}{x \ln x} = \frac{(x+\frac{2}{x})^{\frac{1}{3}}}{x \ln x} > \frac{x^{\frac{1}{3}}}{x \ln x} > \frac{1}{x \ln x}, \quad x \in [2, \infty)$$

$$\therefore \int_2^{\infty} \frac{1}{x \ln x} dx = \int_2^{\infty} \frac{1}{\ln x} d \ln x = \ln(\ln x) \Big|_2^{\infty} = \infty$$

$\therefore \int_2^{\infty} \frac{1}{x \ln x} dx$ is divergent.

$\therefore$ Hence, $\int_2^{\infty} \frac{\sqrt[3]{x^2+2}}{x^3 \ln x} dx$ is divergent by direct comparison. $\square$

Rmk. The direct comparison test also holds for Type-II improper integrals.

The following is another "comparison test".

Theorem (Limit Comparison Test)

Suppose that $a \in \mathbb{R}$, and suppose that f and g are positive continuous functions on $[a, \infty)$. If

"integrable on $[a, b]$, $\forall b \geq a$ "
is enough.

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$$

$\leftarrow f$ and g grow at the same rate.

for some $L \in \mathbb{R}_+ := (0, \infty)$, then

$$\int_a^{\infty} f(x) dx \quad \text{and} \quad \int_a^{\infty} g(x) dx$$

both converge or both diverge.